ASSIGNMENT REPORT

on

Forced Alignment Using Montreal Forced Aligner

Submitted to

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Submitted by:-

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Introduction:-

Alignment in this context refers to the process of synchronizing speech signals with their corresponding transcriptions at both word and phoneme levels. It finds valuable applications in speech recognition, corpus development, and linguistic analysis.

This work implements the MFA framework and analyzes the influence of OOV handling using G2P-based pronunciation modeling.

1. Model and Dictionary Used

- For this project, forced alignment was conducted using the Montreal Forced Aligner in a Kaggle environment. Conda and MFA were installed using the condacolab.

- The pretrained English models, english_us_arpa, were utilized for this purpose. The dataset was copied to the working directory, and the filenames for audio and transcript were converted to lowercase.

- The files were in the correct format, i.e., in MFA format, under the speaker1 folder. Initially, the pre-trained dictionary was used. Words were missing in the dictionary and were identified as Out of Vocabulary words.

- To address OOV words, G2P modeling was employed, and a custom dictionary (custom_dict.txt) was constructed to perform the final alignment

2. Visualization of Sample Alignment

After alignment, Praat was used to create and display TextGrid files.

Figure 1: Example alignment visualization after OOV handling. This figure shows the waveform, spectrogram, and word and phoneme tiers. The boundaries of the alignment match the spoken segments.

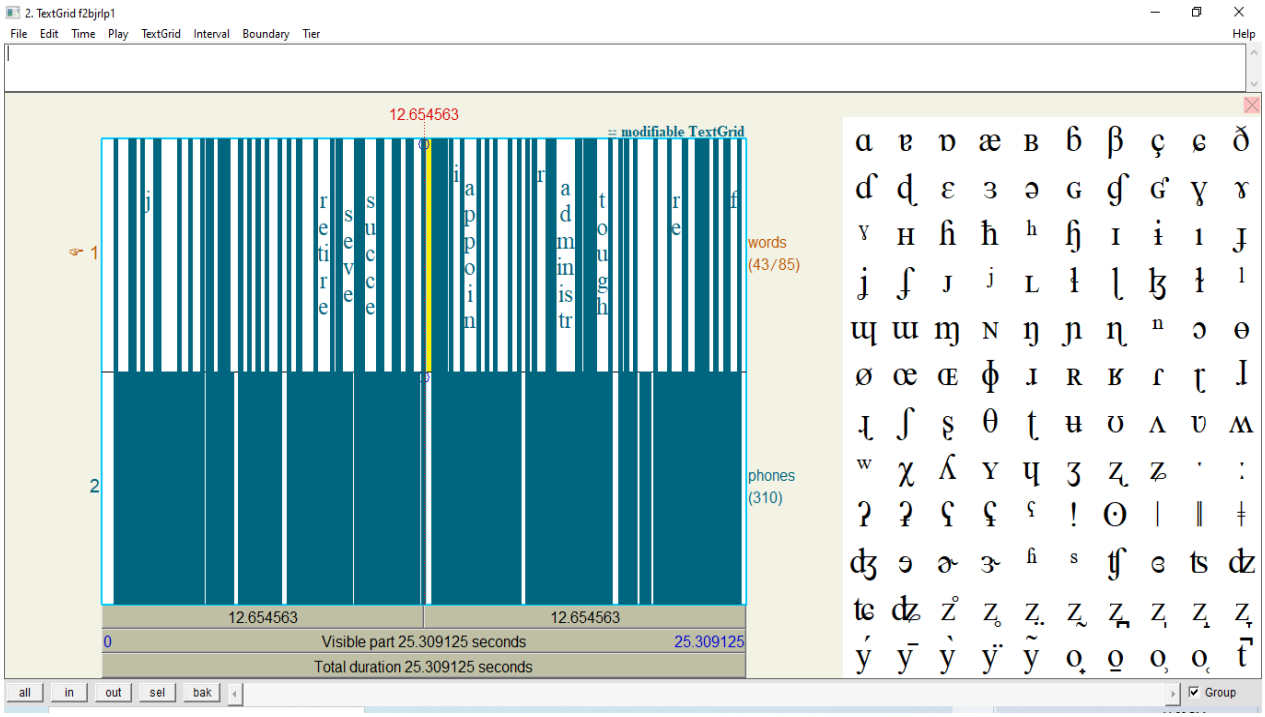

Figure 1: Sample forced alignment visualization after OOV handling using MFA.

3. OOV Handling Before and After: Key Observations

- Two steps of alignments were done: one before OOV handling and the other afterwards.
- Before applying G2P, several words were missing in the dictionary causing alignment errors and also skipped utterances.
- After G2P application and dictionary updates, the alignment accuracy has improved, with a large portion of missing words aligned.

Metric	Before G2P	After G2P
Dictionary Coverage	Limited	Improved
OOV Words	High	Low
Alignment Accuracy	Moderate	High
Skipped Utterances	More	Few

These results indicate that G2P-based OOV handling significantly improved the quality of the alignment.

4. Conclusion

This project successfully implemented forced alignment using MFA with effective OOV handling. G2P-based dictionary enhancement improved pronunciation coverage and resulted in accurate alignment.